Written Assignment 3

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Problem 1

Claim: If $9x + 19$ is even, then x is odd.

Proof:

8: Suppose that x is an integer that is not odd.

5: Therefore, x is even.

9: $\exists k$ such that $x = 2k$.

4: $9x + 19 = 9(2k) + 19 = 18k + 18 + 1$

10: $9x + 19 = 2(9k + 9) + 1$

3: $9x + 19 = 2j + 1$ for some integer j.

7: $9x + 19$ is odd.

Problem 2

Claim: If the sum of two integers is even, then they have the same parity.

Proof:

6: Suppose x and y are integers with opposite parity.

5: Either x is odd and y is even or x is even and y is odd.

4: Without loss of generality assume x is even and y is odd.

3: $\exists k, j \in \mathbb{Z}$ such that $x = 2k$ and $y = 2j + 1$

7: $x + y = 2k + 2j + 1$

9: $\exists s$ such that $x + y = 2s + 1$

2: $x + y$ is odd.

8: Therefore, x + y is even $\rightarrow$ x and y have the same parity.